

```

1  #include<stdio.h>
2  #include<string.h>
3  #include<fcntl.h>
4  #include<unistd.h>
5  #include<sys/stat.h>
6  #define FIFO_NAME "myfifo"
7  int main(){
8      int fd;
9      char msg[100];
10     mkfifo(FIFO_NAME, 0666);
11     fd = open(FIFO_NAME, O_WRONLY);
12     printf("Enter data: ");
13     fgets(msg, sizeof(msg), stdin);
14     write(fd, msg, strlen(msg) + 1);
15     close(fd);
16 }
17 #####
18 #include<stdio.h>
19 #include<fcntl.h>
20 #include<unistd.h>
21 #include<sys/stat.h>
22 #define FIFO_NAME "myfifo"
23 int main(){
24     int fd;
25     char msg[100];
26     fd = open(FIFO_NAME, O_RDONLY);
27     read(fd, msg, sizeof(msg));
28     printf("Received: %s", msg);
29     close(fd);
30 }
31 #####
32 #include <pthread.h>
33 #include <semaphore.h>
34 #include <stdio.h>
35 #include <unistd.h>
36 int buffer[5];
37 int front = 0, rear = 0;
38 sem_t mutex, full, empty;
39 void *producer(void *arg) {
40     for (int i = 0; i < 10; i++) {
41         sem_wait(&empty);
42         sem_wait(&mutex);
43         rear = (rear + 1) % 5;
44         buffer[rear] = i;
45         printf("Produced: %d\n", i);
46         sleep(1);
47         sem_post(&mutex);
48         sem_post(&full);
49     }
50     return NULL;
51 }
52
53 void *consumer(void *arg) {
54     int item;
55     for (int i = 0; i < 10; i++) {
56         sem_wait(&full);
57         sem_wait(&mutex);
58         front = (front + 1) % 5;
59         item = buffer[front];
60         printf("Consumed: %d\n", item);
61         sleep(1);
62         sem_post(&mutex);
63         sem_post(&empty);
64     }
65     return NULL;
66 }
67 int main() {
68     pthread_t producer_thread, consumer_thread;
69     sem_init(&mutex, 0, 1);
70     sem_init(&full, 0, 0);
71     sem_init(&empty, 0, 5);
72     pthread_create(&producer_thread, NULL, producer, NULL);
73     pthread_create(&consumer_thread, NULL, consumer, NULL);
74     pthread_join(producer_thread, NULL);
75     pthread_join(consumer_thread, NULL);
76 }

```